Ruo-Yu Guan

Ph.D. Candidate in Theoretical Physics
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Research Interests

- Space-based gravitational-wave astrophysics.
- Galactic compact binaries, particularly double white dwarfs.
- Stochastic gravitational-wave backgrounds, foreground statistics, and Bayesian inference.

Education

Ph.D. in Physics (Theoretical Physics) Sep. 2025 – Jun. 2029 (expected)
School of Physics, HUST *Wuhan, China*
Doctoral coursework completed; thesis proposal successfully defended.
Ranked 2nd of 138 Ph.D. students based on course grades.
Supervisor: Prof. Yan Wang

M.Sc. in Physics Nov. 2024
Department of Physics, The University of Hong Kong (HKU) *Hong Kong SAR, China*
Capstone Project: *Identifying Sources in Cygnus OB2 Using Multiwavelength Observations.*
Supervisor: Assoc. Prof. Stephen Chi-Yung Ng

B.Sc. Jun. 2022
Jilin University *Changchun, China*
Graduation Thesis: *Study on the Prompt Emission of Gamma-Ray Bursts and Its Polarization.*
Research Mentor: Prof. Mi-Xiang Lan

Research Positions

Research Assistant Nov. 2024 – Jun. 2025
Department of Astronomy, HUST *Wuhan, China*
Research Supervisor: Prof. Yuan-Chuan Zou

Publications

[1] **Ruo-Yu Guan** and Yan Wang. Residual Galactic binary foreground in LISA stochastic gravitational-wave background inference: Source power concentration and spectral degeneracy. *Physical Review D*, accepted (2026). doi:10.1103/4779-jw6l.

[2] **Ruo-Yu Guan**, Fei-Fei Wang, and Yuan-Chuan Zou. Hurst index of gamma-ray burst light curves and its statistical study. *Journal of High Energy Astrophysics*, **51**, 100559 (2026). doi:10.1016/j.jheap.2026.100559.

[3] **Ruo-Yu Guan** and Mi-Xiang Lan. Interpreting time-integrated polarization data of gamma-ray burst prompt emission. *Astronomy & Astrophysics*, **670**, A160 (2023). doi:10.1051/0004-6361/202243805.

Research Experience

Galactic-Binary Gravitational-Wave Foregrounds
National Gravitation Laboratory and School of Physics, HUST *Jul. 2025 – Present*

- Investigating the angular anisotropy of the unresolved Galactic-binary gravitational-wave foreground.

Research Experience (continued)

Residual Galactic Binary Foregrounds in LISA SGWB Inference
National Gravitation Laboratory and School of Physics, HUST *2026*

- Estimated the residual Galactic-binary power spectrum using injection–recovery catalogs.
- Quantified non-Gaussianity from source discreteness using excess kurtosis. Instrumental noise and SGWB power largely suppress this signature.
- Assessed spectral degeneracy with an isotropic SGWB using Fisher analysis. Marginalizing over the residual-power normalization increased the SGWB-amplitude uncertainty by 13.6% in the fiducial Gaussian-covariance model.

Time-Series and Correlation Analysis of Gamma-Ray Burst Light Curves
Department of Astronomy, HUST *Oct. 2024 – Jan. 2026*

- Measured Hurst indices for 163 long-duration BATSE gamma-ray bursts using detrended fluctuation analysis.
- Found anti-correlations with burst durations and moderate positive correlations with peak photon flux. Standard spectral parameters showed no significant linear correlations.

Identifying Sources in Cygnus OB2 Using Multiwavelength Observations
Department of Physics, HKU *Sep. 2023 – Jun. 2024*

- Analyzed Chandra X-ray observations of Cygnus OB2 and cross-matched the detected sources with Gaia.
- Identified optical counterparts for 106 of 148 X-ray sources. Hardness ratios suggested that 12 unmatched sources might be background galaxies.

Gamma-Ray Burst Prompt Emission and Polarization
Center for Theoretical Physics and College of Physics, Jilin University *2022*

- Predicted upper limits on time-integrated polarization for 37 GRBs using synchrotron emission in ordered magnetic fields.
- Most polarization observations were consistent with the model, while two bursts exceeded the predicted limits.
- Predicted energy-dependent polarization signatures for POLAR-2 to test competing prompt-emission models.

Conference Presentations

Residual Galactic binary foreground in LISA stochastic gravitational-wave background inference
Shanghai International Symposium on Relativistic Astrophysics
Poster presentation Shanghai, China Dec. 2026

Galactic Binary Residuals in LISA SGWB Amplitude Inference: Source Power Concentration and Spectral Degeneracy
National Key R&D Program Mid-Year Workshop | Oral presentation Yangzhou, China Jul. 2026

Time-integrated Polarizations of Gamma-ray Burst Prompt Phase
PKU International PhD Student Forum | Poster presentation Beijing, China Dec. 2024

Academic Memberships

LISA Consortium Mar. 2026 – Present
Core Member, Astrophysics Working Group

Laboratory for Space Research Oct. 2023 – Nov. 2024
Member, The University of Hong Kong

Skills

Research Methods: Statistical analysis and inference (Fisher-matrix analysis, correlation analysis); time-series analysis; binary population synthesis (COSMIC).

Programming & Tools: Python (NumPy, SciPy, pandas, Matplotlib); MATLAB, Mathematica, R (familiar); Git, Linux, LaTeX.

Languages: Chinese (Mandarin: native; Cantonese: conversational); English (English-taught M.Sc. at HKU).